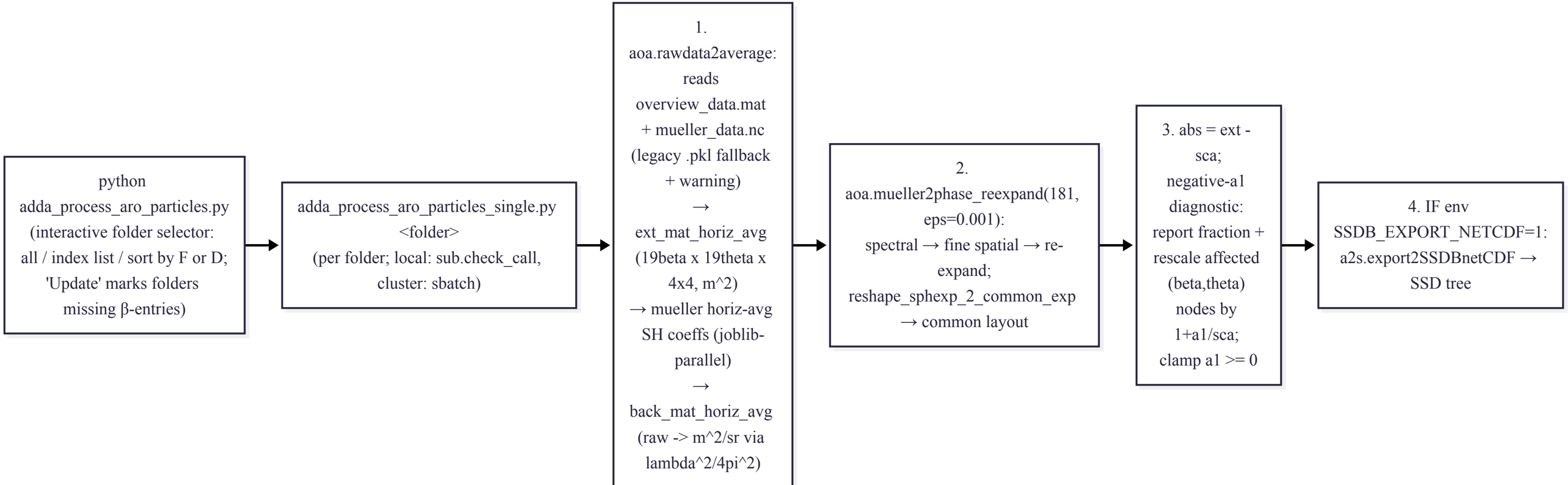

python adda_process_aro_particles.py (interactive folder selector: all / index list / sort by F or D; 'Update' marks folders missing β-entries)
adda_process_aro_particles_single.py <folder> (per folder; local: sub.check_call, cluster: sbatch)
1. aoa.rawdata2average: reads overview_data.mat + mueller_data.nc (legacy .pkl fallback + warning) → ext_mat_horiz_avg (19beta x 19theta x 4x4, m^2) → mueller horiz-avg SH coeffs (joblib-parallel) → back_mat_horiz_avg (raw -> m^2/sr via lambda^2/4pi^2)
2. aoa.mueller2phase_reexpand(181, eps=0.001): spectral → fine spatial → re-expand; reshape_sphexp_2_common_exp → common layout
3. abs = ext - sca; negative-a1 diagnostic: report fraction + rescale affected (beta,theta) nodes by 1+a1/sca; clamp a1 >= 0
4. IF env SSDB_EXPORT_NETCDF=1: a2s.export2SSDBnetCDF → SSD tree